

Date:

EXPERIMENT NO. 6

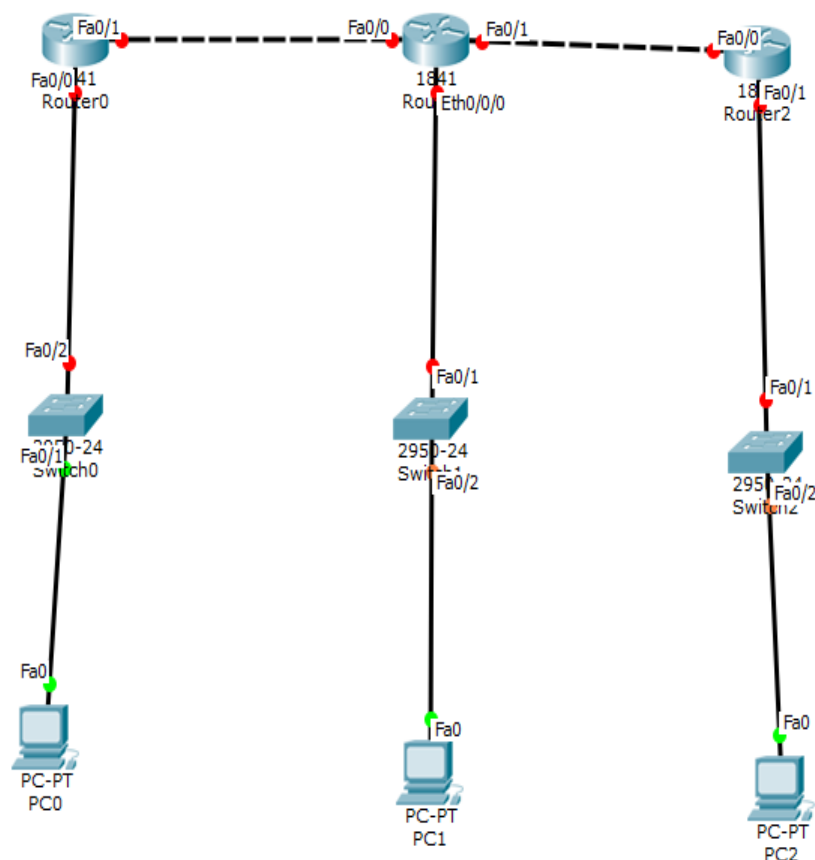
AIM: Understand & identify Packet(L3) & frame(L2) content detail.

Tools required:

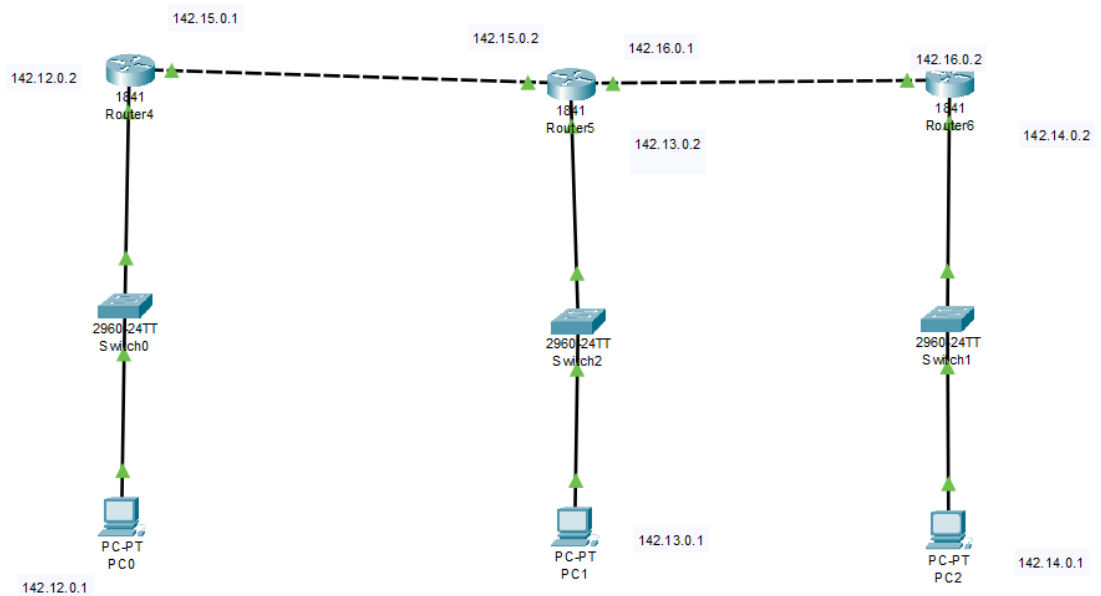
1. Desktop Computer
2. Cisco Packet Tracer

Simulate the different scenarios given below in the Cisco packet tracker and fill up the table.

Note: While applying an IP address, the student needs to allocate an IP address as per his/her student ID. For Example, if the student ID is 22cs005 then IP address allocation for the first network should start with 5.0.0.0. For subsequent networks, it should start with ID+1 i.e. 6.0.0.0, 7.0.0.0 and so on.



Topology for the consideration



Steps:

1. Create topology in Realtime mode
2. Configure IP address
3. Configure Dynamic Routing in Each routers
4. Ping from PC0 to rest other PCs and all interface of routers and fill success table
5. Go to Simulation mode
6. Prepare MAC and IP address Table.
7. Prepare ARP table for all PCs
8. Prepare Routing tables for Router0, Router1 and Router2
9. Prepare ARP tables for Router0, Router1 and Router2
10. Prepare MAC table of all switches
11. Ping from PC0 to PC2 and attach a snapshot for the same.
12. Insert screenshots of all PCs, Switches and Routers for ARP table, MAC table and Routing table.

In simulation mode**Exercise-1:** Ping from PC0 to FE0/0 of Router0

click on capture forward once so packet goes to switch, Inspect&
insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU

PDU Information at Device: Switch0

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetII

0		4		8		Bytes	
PREAMBLE: 101010...10				SF D	DEST ADDR:000C.85D6.B4 01		
SRC ADDR:0060. 3E6D.28DE		TYPE:0 x0800		DATA (VARIABLE LENGTH)		FCS:0x00000000	

IP

0		4		8		16		20		24		Bits	
VER:4		IHL:5		DSCP:0x00		TL:28							
ID:0x0005				FLAGS: 0x0		FRAG OFFSET:0x000							
TTL:255		PRO:0x01		CHKSUM									
SRC IP:142.12.0.1													
DST IP:142.14.0.1													
DATA (VARIABLE LENGTH)													

ICMP

OUTBOUND PDU

PDU Information at Device: Switch0

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetII

0		4		8		Bytes	
PREAMBLE: 101010...10				SF D	DEST ADDR:000C.85D6.B4 01		
SRC ADDR:0060. 3E6D.28DE		TYPE:0 x0800		DATA (VARIABLE LENGTH)		FCS:0x00000000	

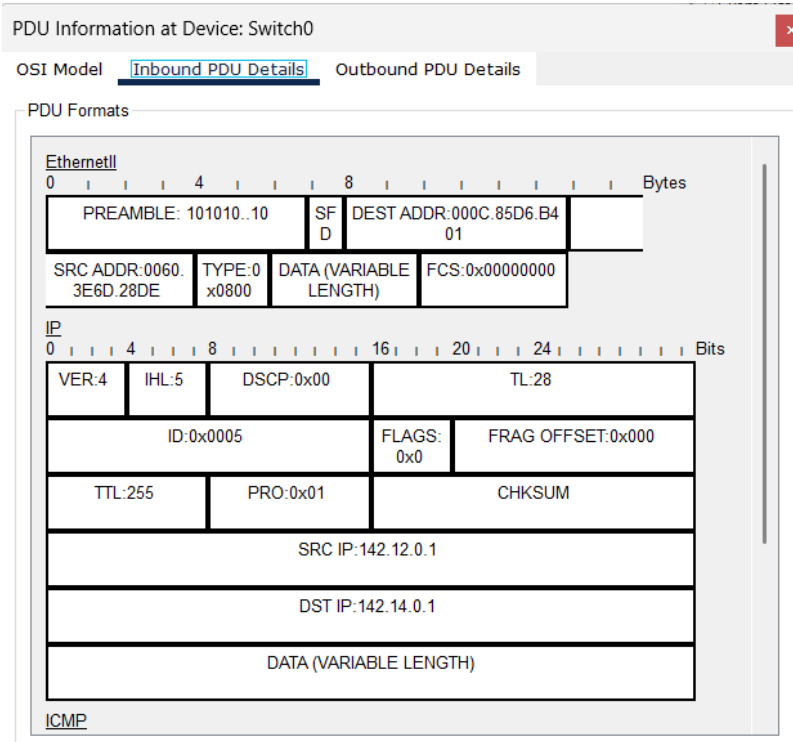
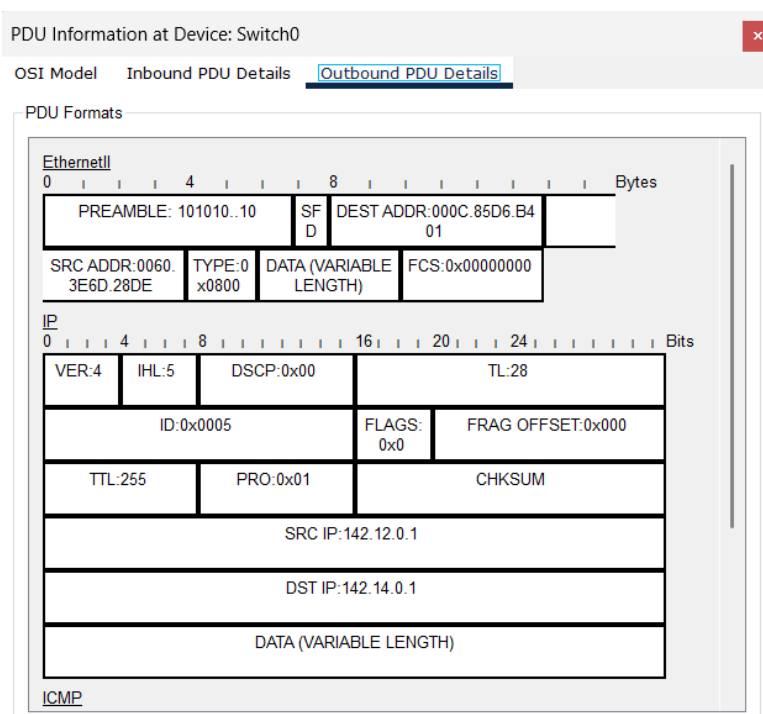
IP

0		4		8		16		20		24		Bits	
VER:4		IHL:5		DSCP:0x00		TL:28							
ID:0x0005				FLAGS: 0x0		FRAG OFFSET:0x000							
TTL:255		PRO:0x01		CHKSUM									
SRC IP:142.12.0.1													
DST IP:142.14.0.1													
DATA (VARIABLE LENGTH)													

ICMP

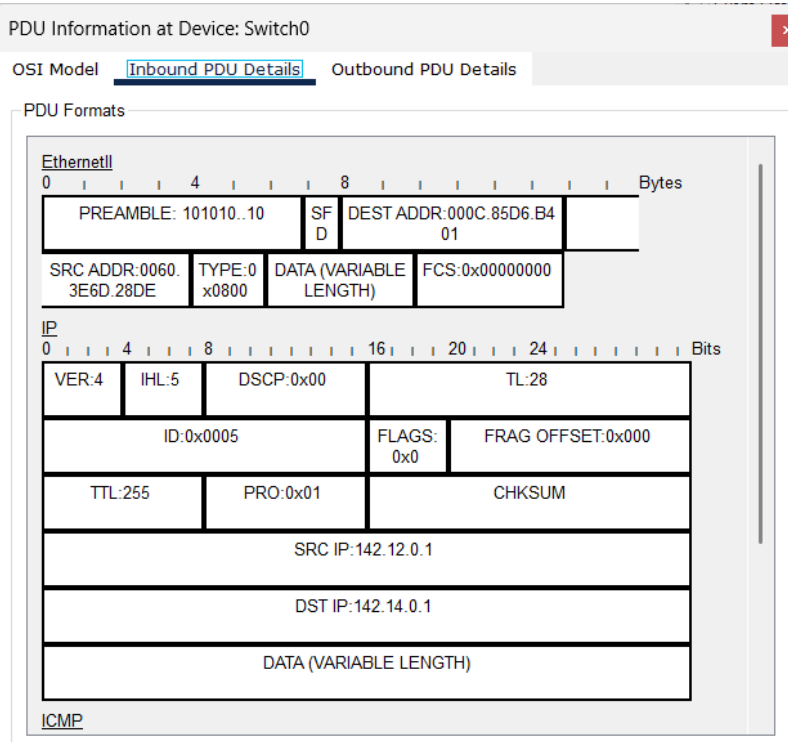
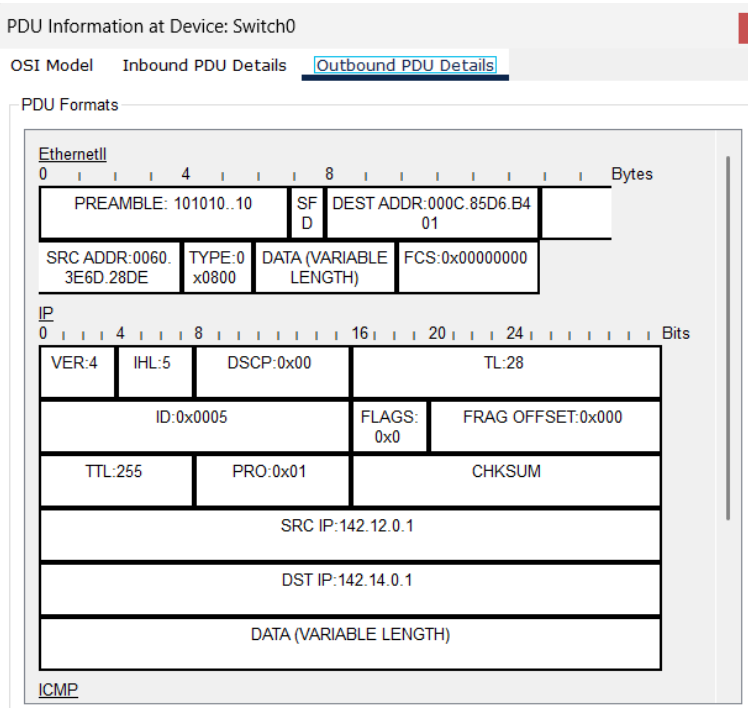
Exercise-2: Ping from PC0 to FE0/1 of Router0

Click on capture forward once so packet goes to switch, Inspect& insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU**OUTBOUND PDU**

Exercise-3: Ping from PC0 to FE0/0 of Router1

Click on capture forward once so packet goes to switch, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU**OUTBOUND PDU**

INBOUND PDU

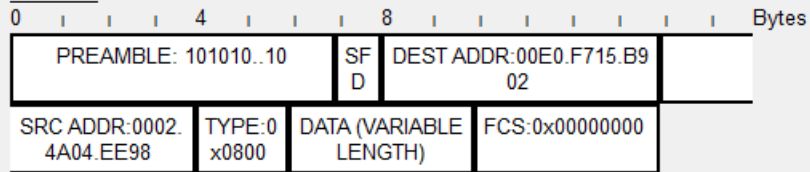
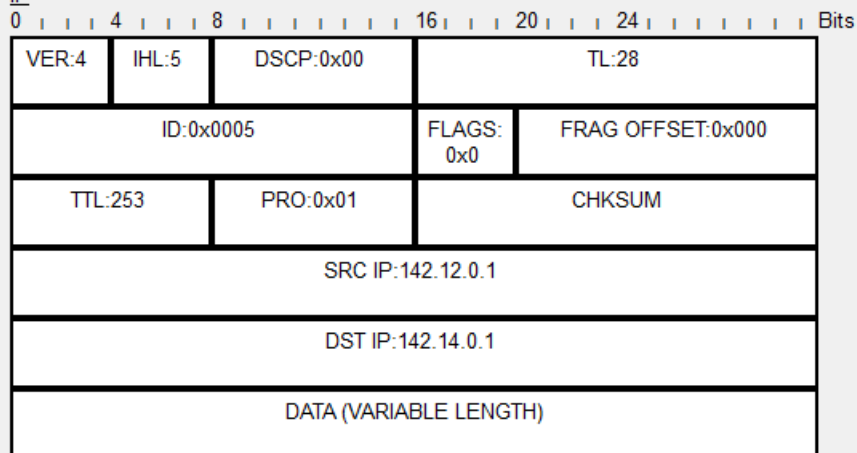
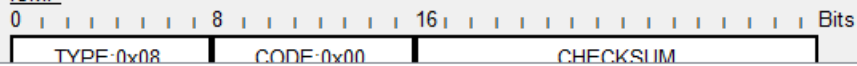
Outbound PDU Details

0 8 16 Bits

TYPE: 0x08 CODE: 0x00 CHECKSUM

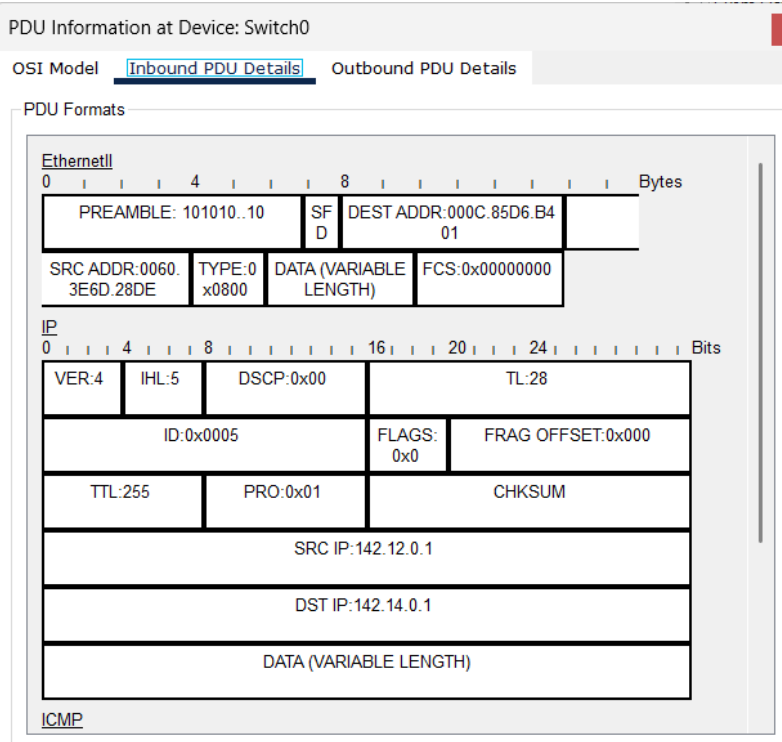
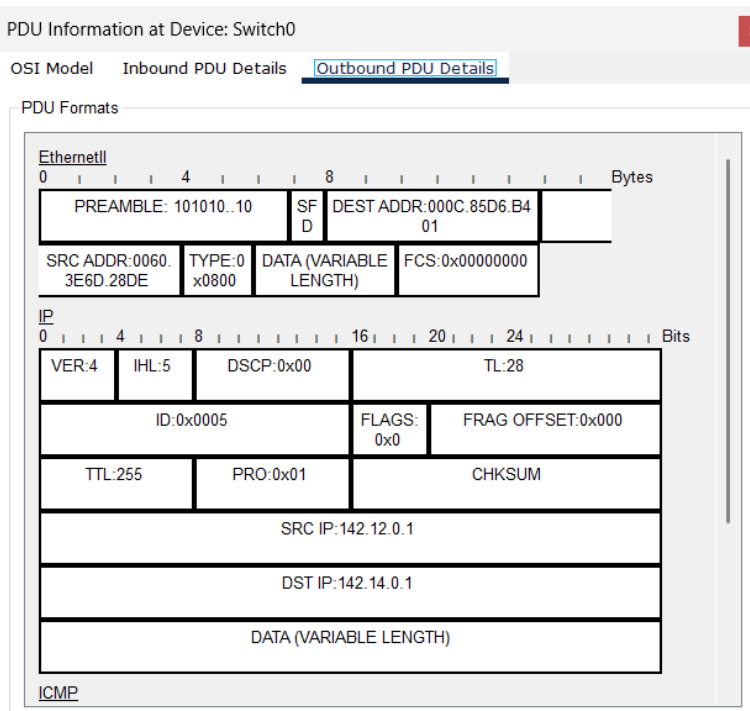
OUTBOUND PDUOSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetIIIPICMP

Exercise-4: Ping from PC0 to PC1 (For even roll number of student PC0 to PC2)

Click on capture forward so packet goes to switch0, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU**OUTBOUND PDU**

Click on capture forward so packet goes to Router0, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU[OSI Model](#)[Inbound PDU Details](#)[Outbound PDU Details](#)

PDU Formats

EthernetII

0				4				8									Bytes
PREAMBLE: 101010..10								SF D	DEST ADDR:0002.17A3.72 01								
SRC ADDR:000C. 85D6.B402				TYPE:0 x0800		DATA (VARIABLE LENGTH)				FCS:0x00000000							

IP

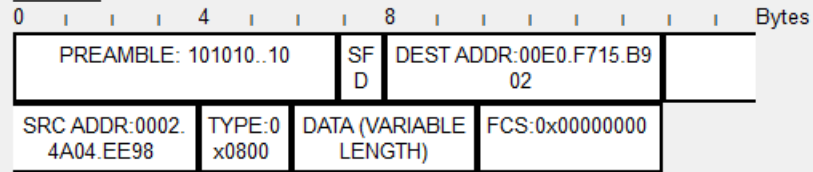
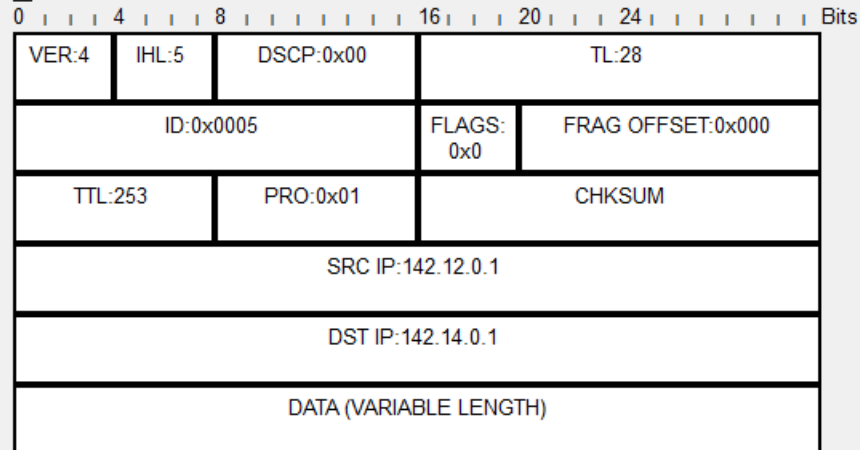
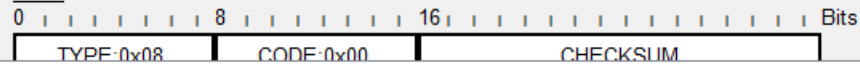
0 4 8 16 20 24 Bits																															
VER:4				IHL:5				DSCP:0x00								TL:28															
ID:0x0005																FLAGS: 0x0				FRAG OFFSET:0x000											
TTL:254								PRO:0x01								CHKSUM															
SRC IP:142.12.0.1																															
DST IP:142.14.0.1																															
DATA (VARIABLE LENGTH)																															

ICMP

0								8									16												Bits
TYPE:0x08								CODE:0x00								CHECKSUM													

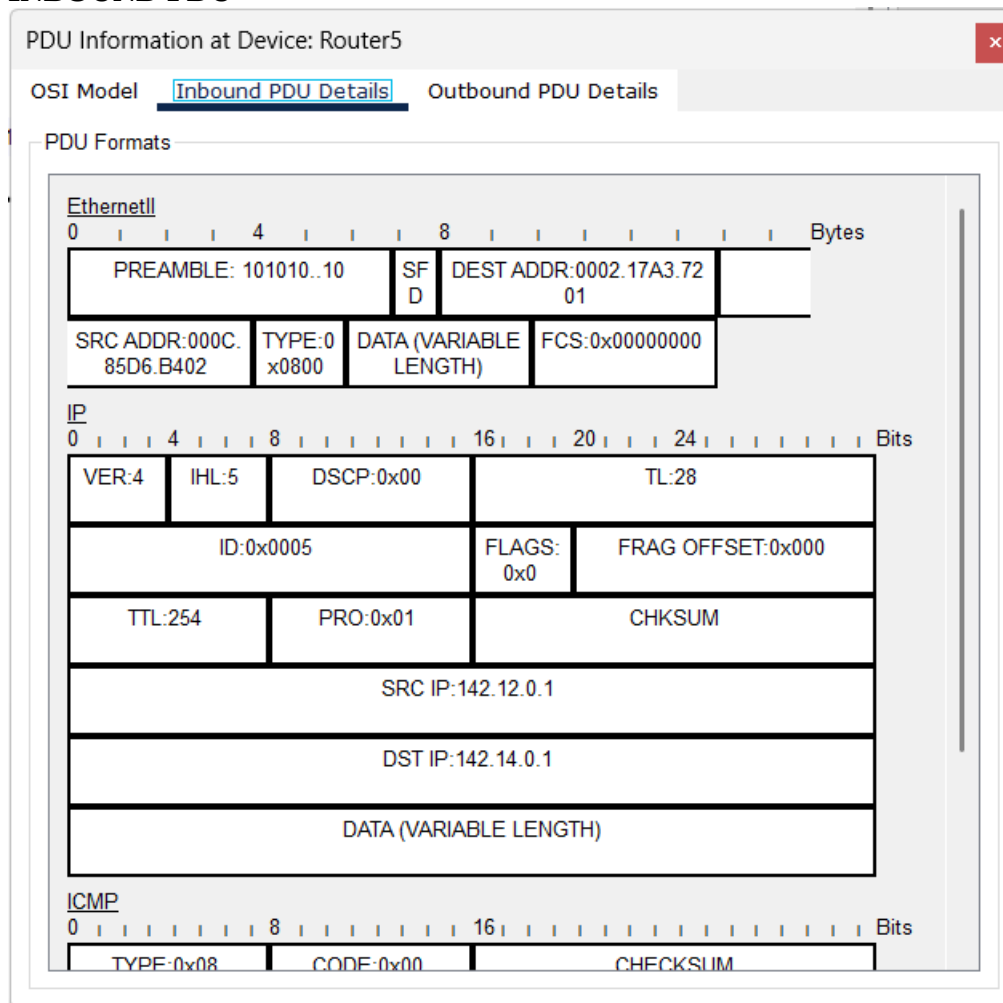
OUTBOUND PDUOSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetIIIPICMP

Click on capture forward so packet goes to Router1, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU



OUTBOUND PDU

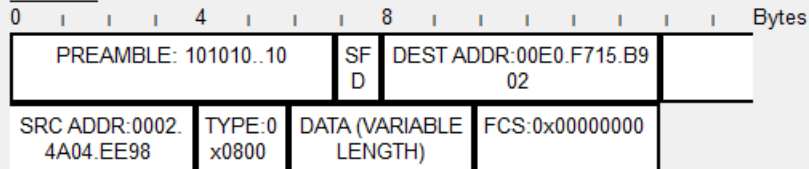
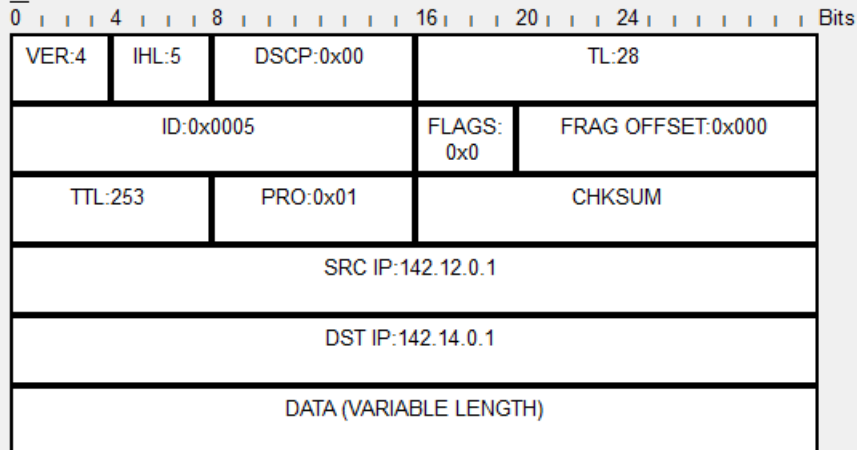
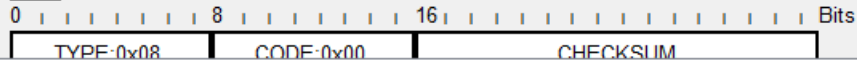
PDU Information at Device: Router5

OSI Model

Inbound PDU Details

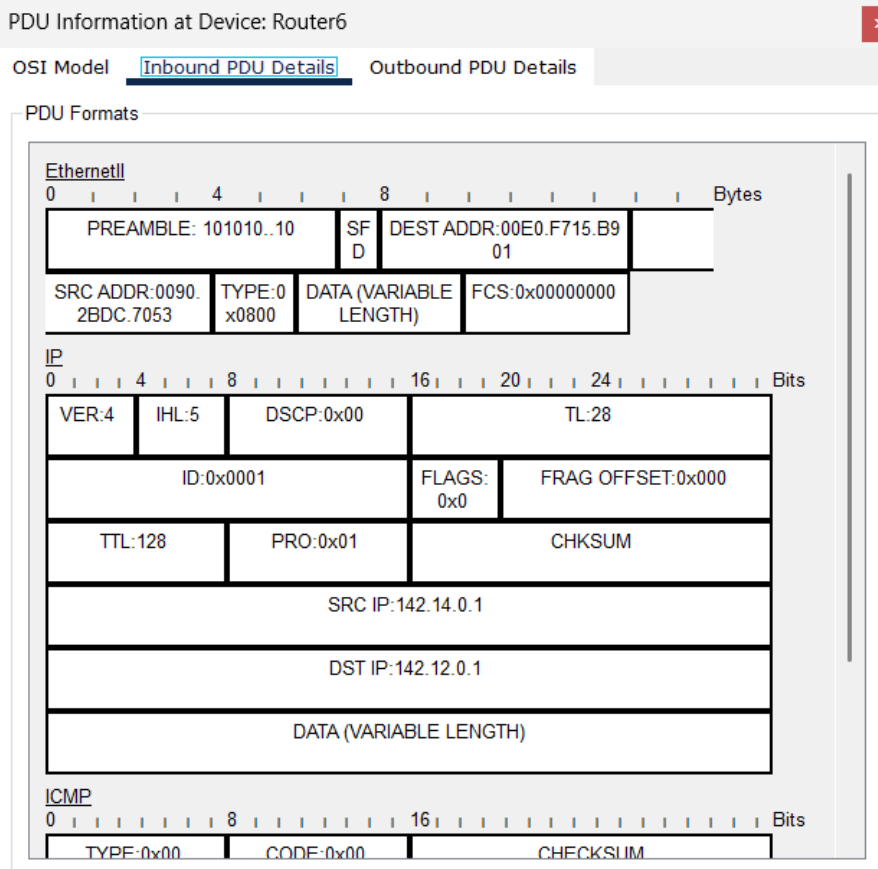
Outbound PDU Details

PDU Formats

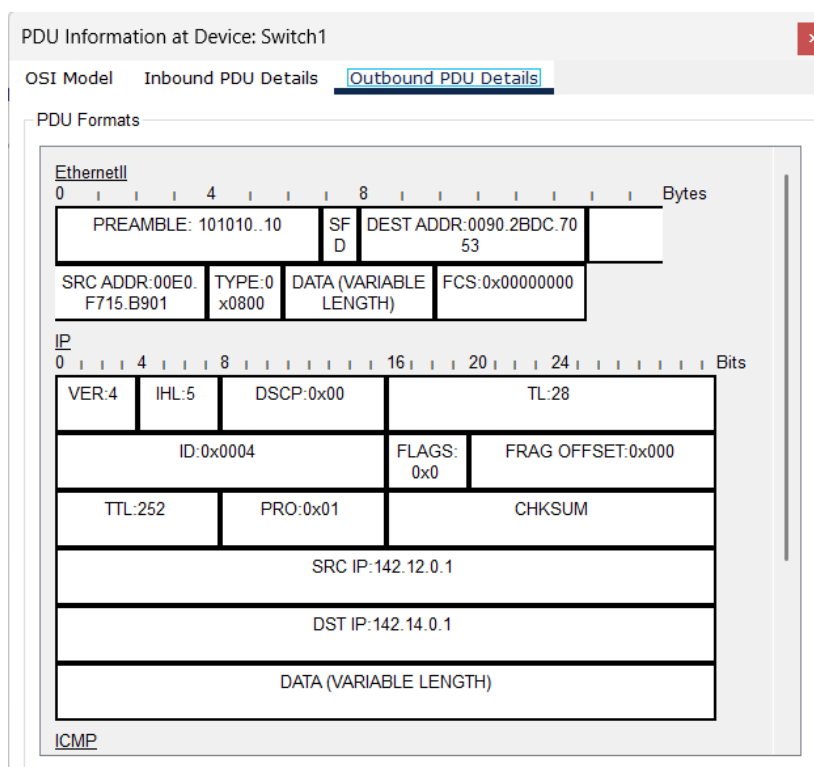
EthernetIIIPICMP

Click on capture forward so packet goes to switch2, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU



OUTBOUND PDU



Click on capture forward so packet goes to PC1, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU

PDU Information at Device: PC2

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetII

0 4 8 Bytes			
PREAMBLE: 101010..10		SF D	DEST ADDR:0090.2BDC.7053
SRC ADDR:00E0.F715.B901	TYPE:0x0800	DATA (VARIABLE LENGTH)	FCS:0x00000000

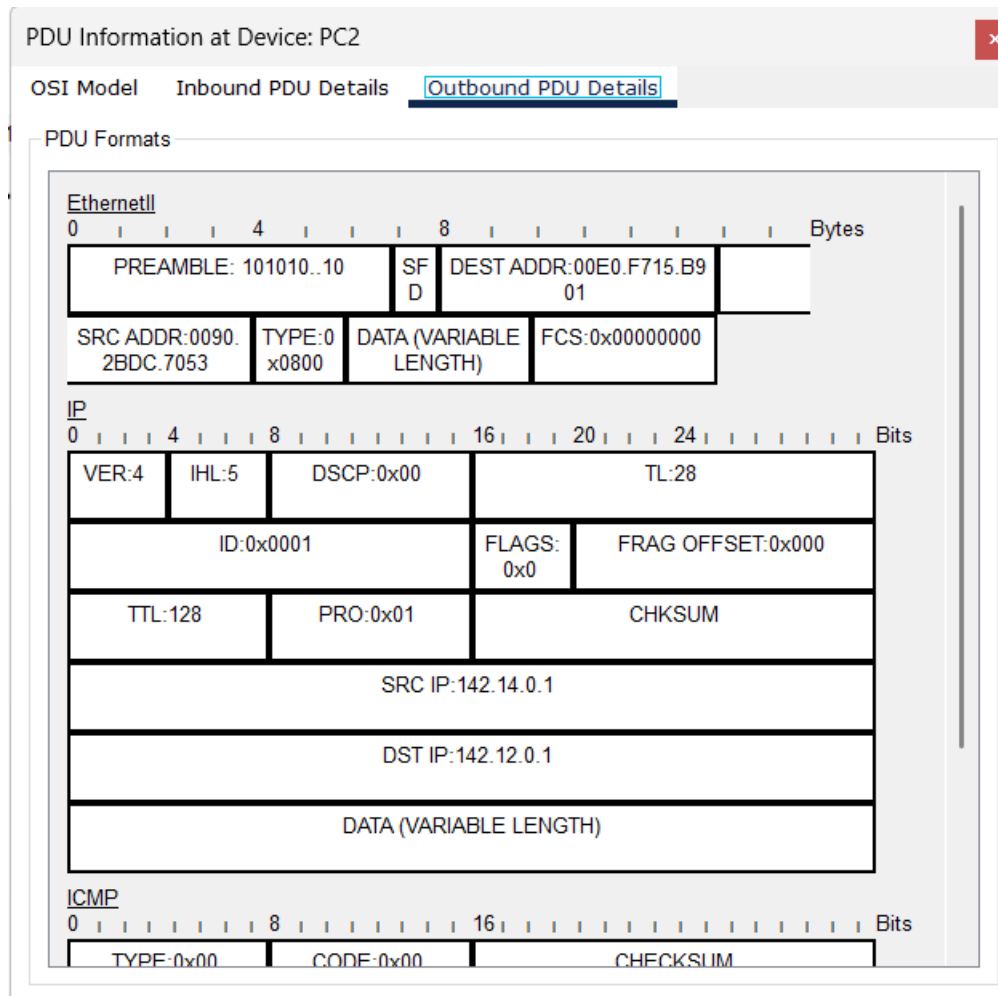
IP

0 4 8 16 20 24 Bits			
VER:4	IHL:5	DSCP:0x00	TL:28
ID:0x0004		FLAGS:0x0	FRAG OFFSET:0x000
TTL:252	PRO:0x01	CHKSUM	
SRC IP:142.12.0.1			
DST IP:142.14.0.1			
DATA (VARIABLE LENGTH)			

ICMP

0 8 16 Bits		
TYPE:0x08	CODE:0x00	CHECKSUM

OUTBOUND PDU



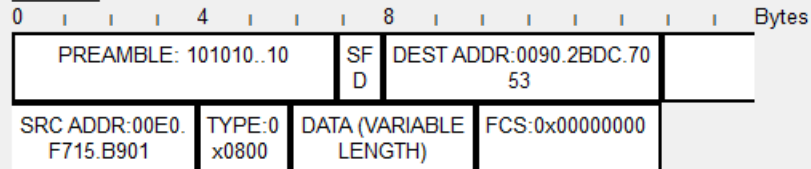
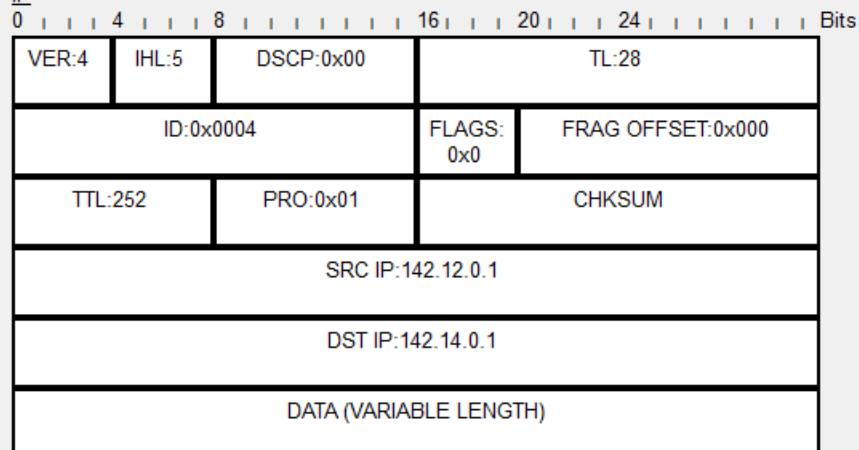
Click on capture forward so packet goes to switch1, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

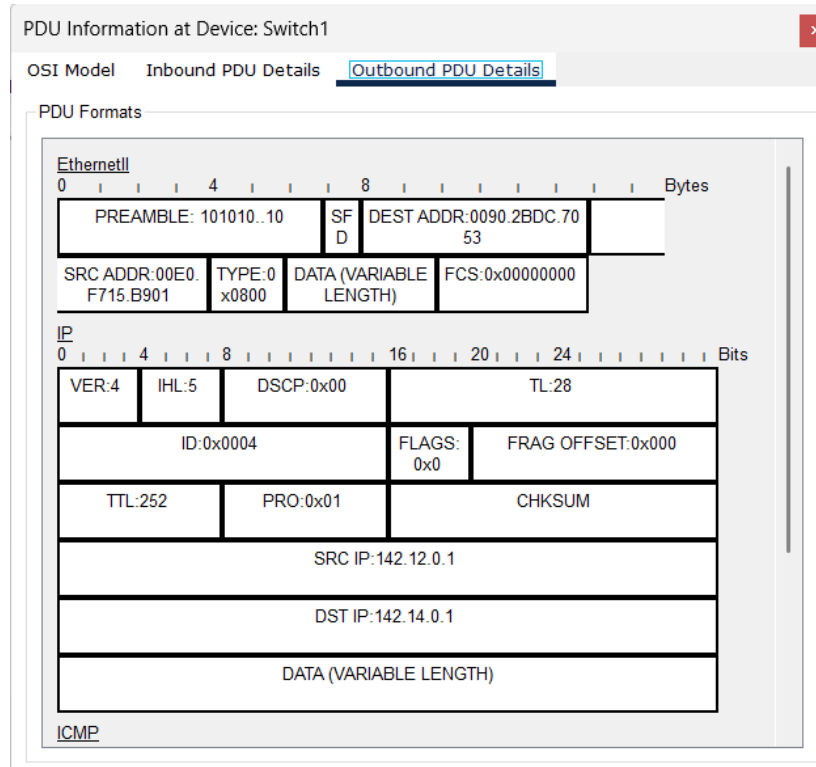
INBOUND PDU

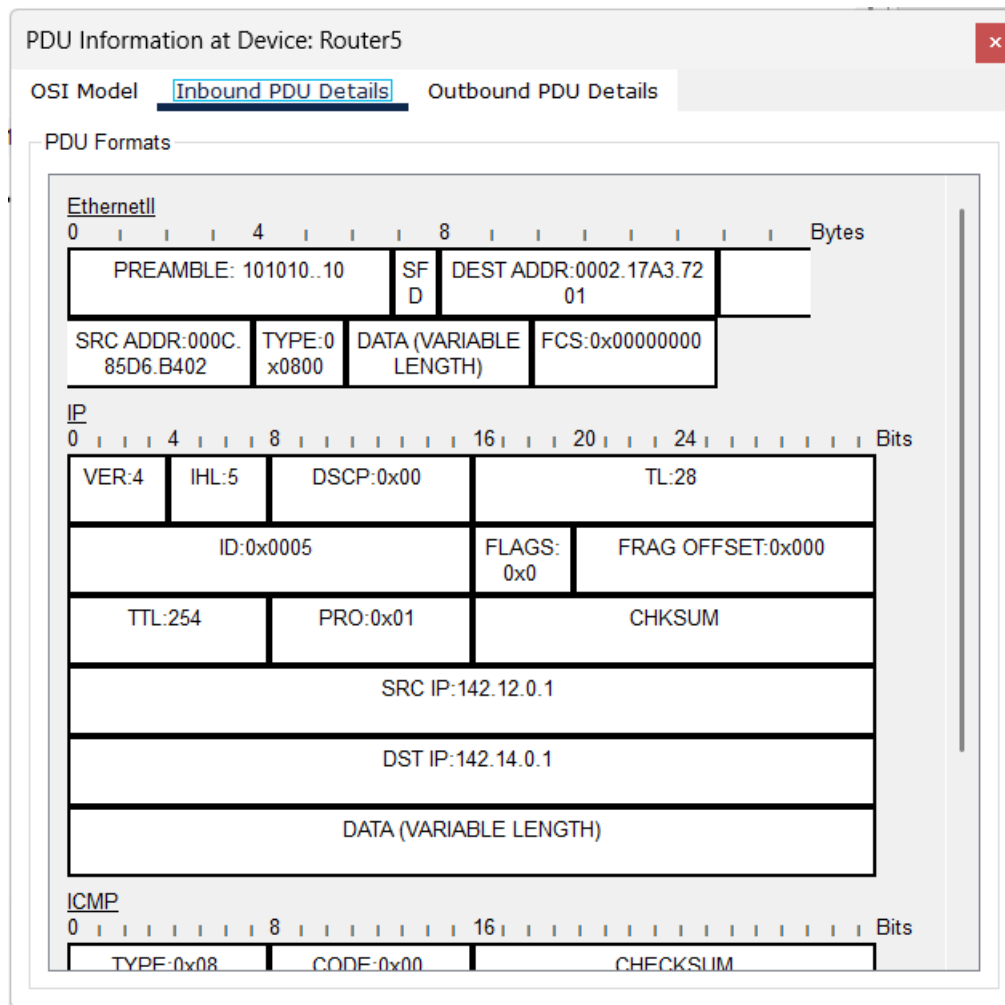
PDU Information at Device: Switch1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

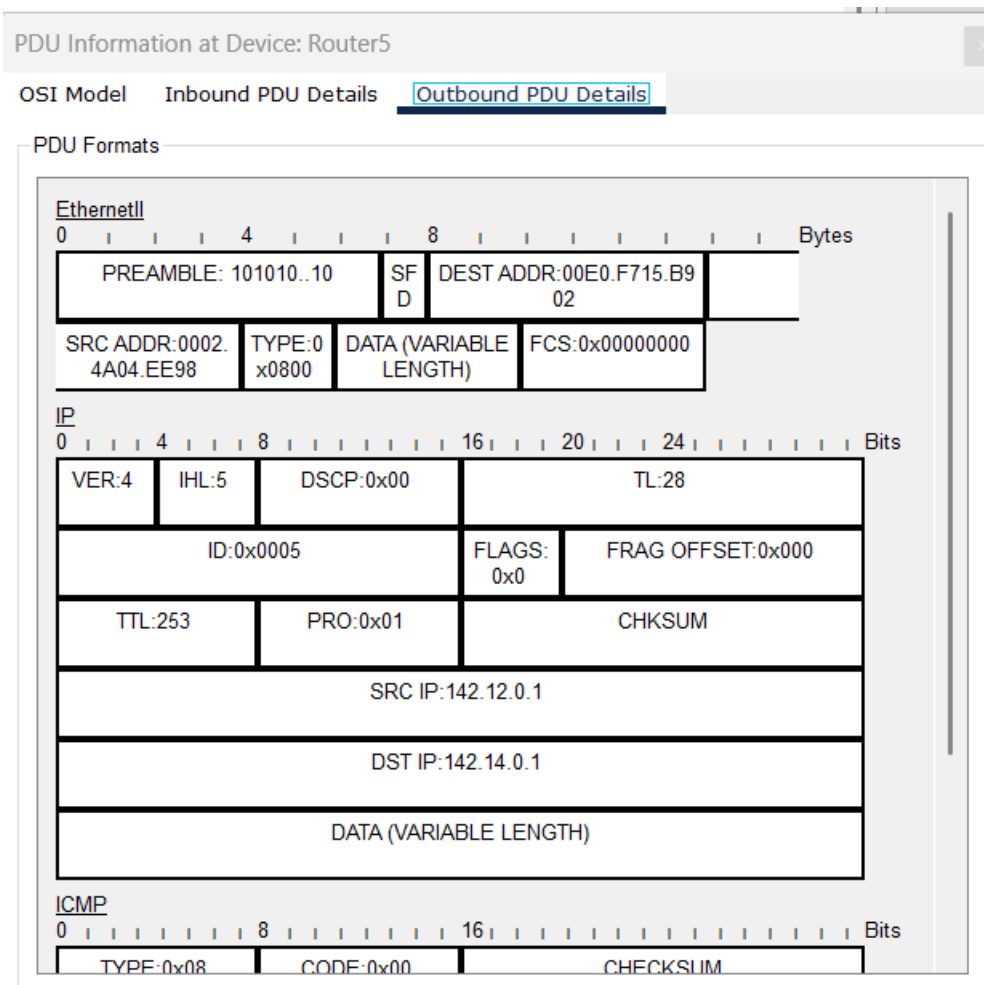
PDU Formats

EthernetIIIPICMP



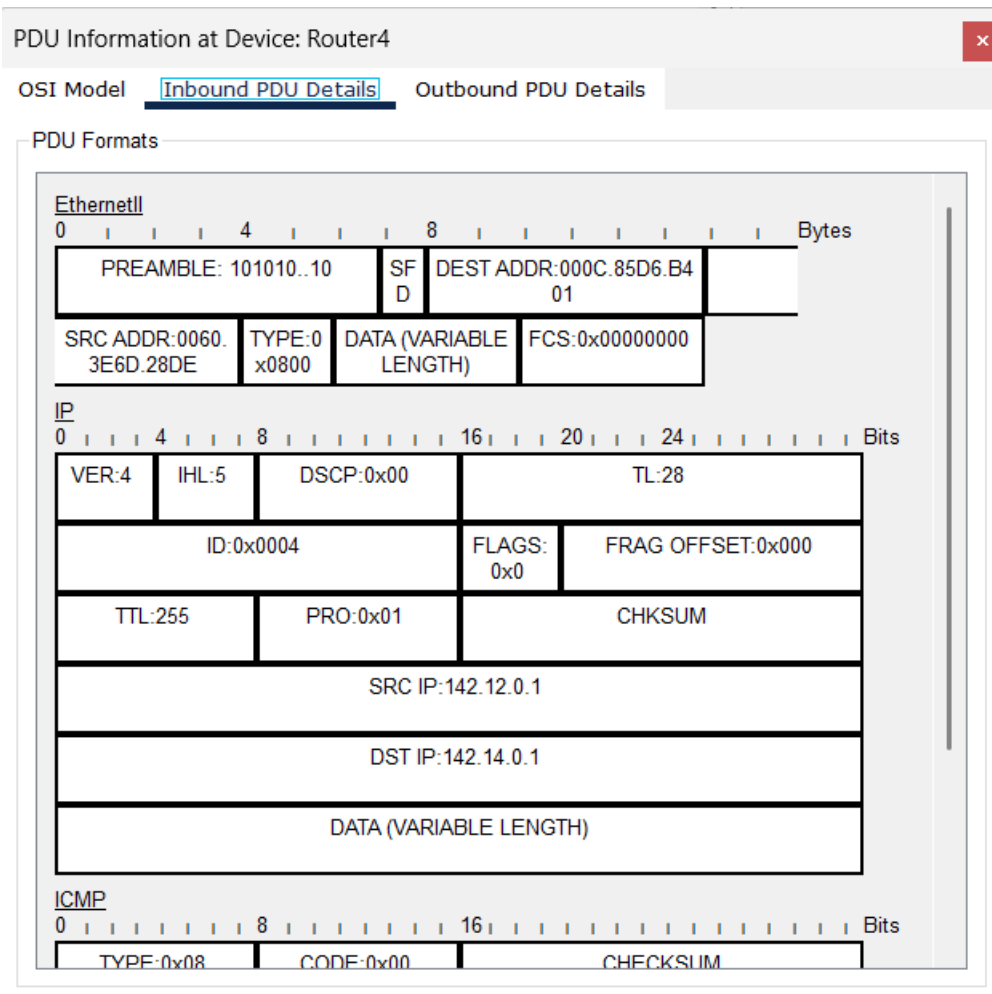


OUTBOUND PDU



Click on capture forward so packet goes to router0, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU



OUTBOUND PDU

PDU Information at Device: Router4

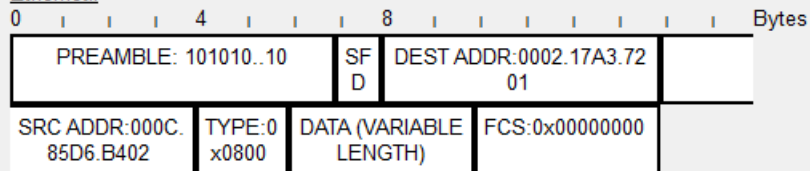
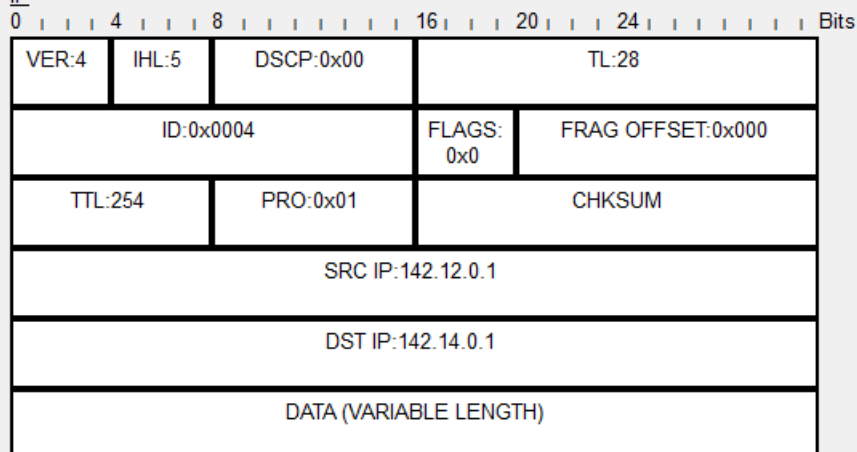
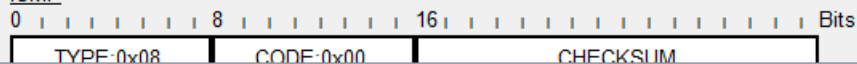


OSI Model

Inbound PDU Details

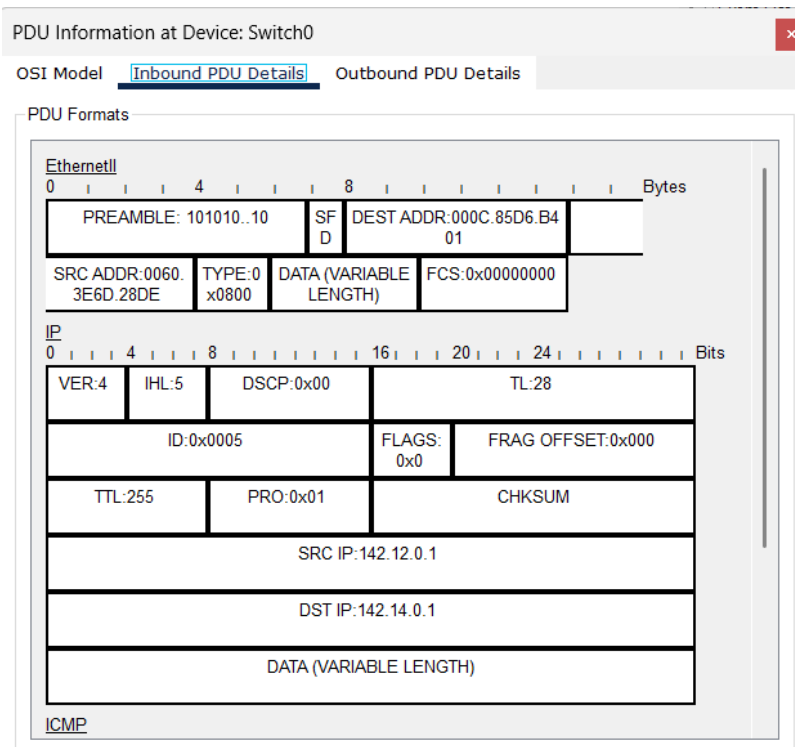
Outbound PDU Details

PDU Formats

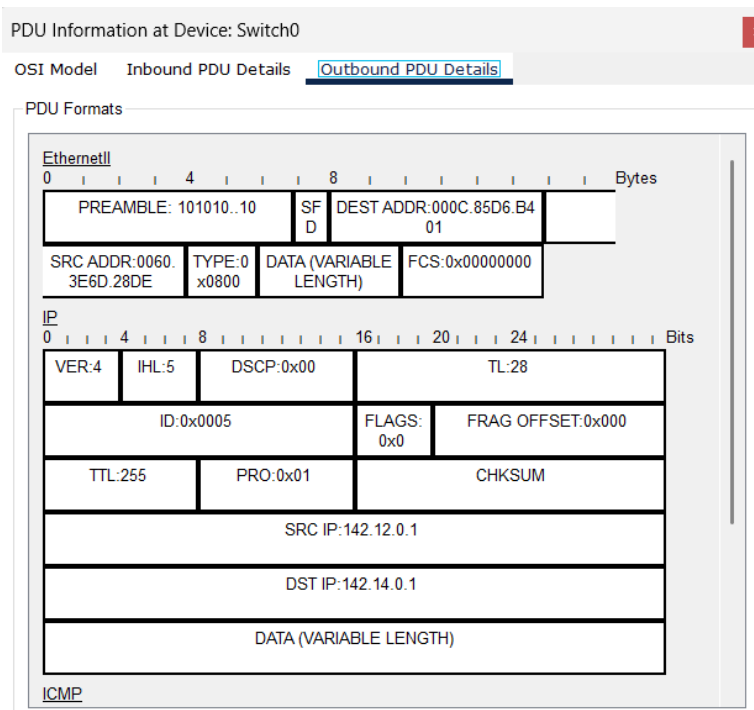
EthernetIIIPICMP

Click on capture forward so packet goes to switch0, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU

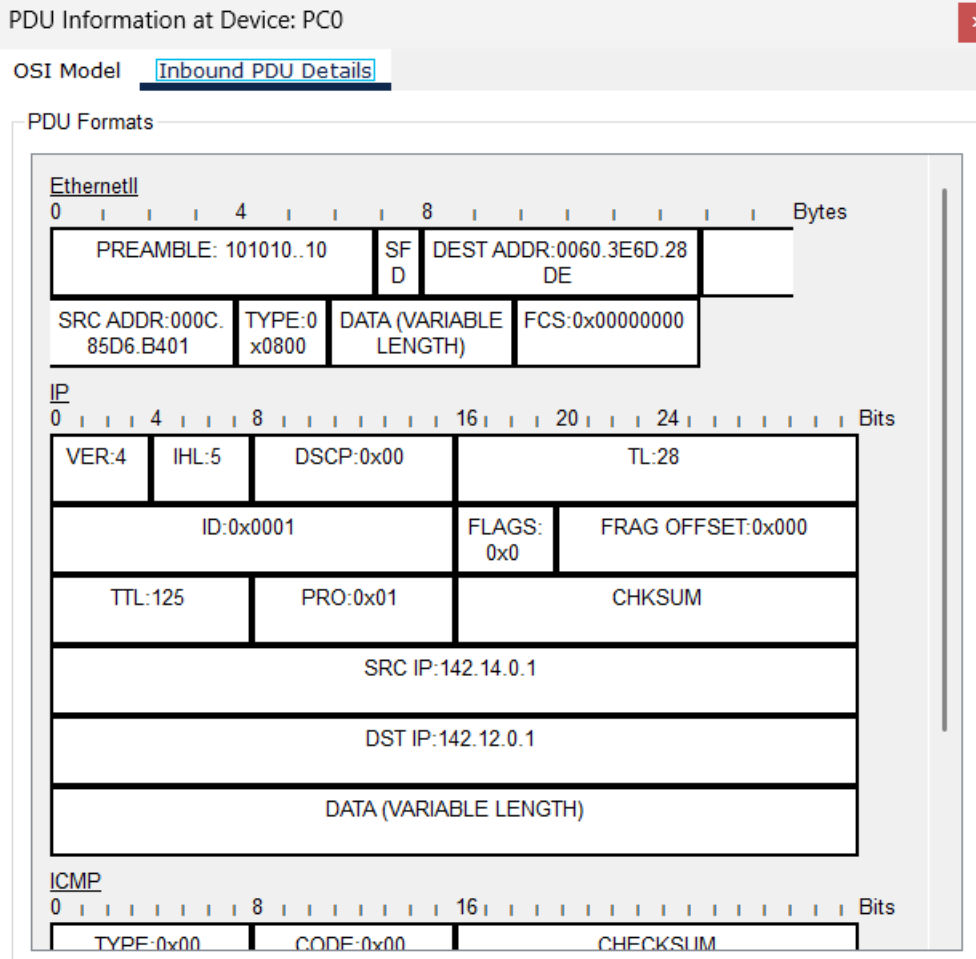


OUTBOUND PDU



Click on capture forward so packet goes to PC0, Inspect & insert screenshots of Inbound and Outbound PDU and fill following table

INBOUND PDU



OUTBOUND PDU

NO Outbound pdu is present